

State estimation

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Control y Sistemas

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Universidad Nacional de Cuyo

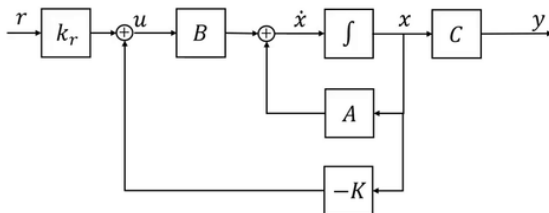
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State feedback control

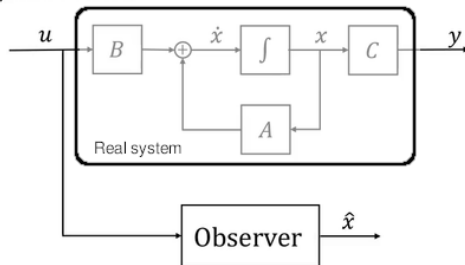
Idea with state feedback control design: Modify the eigenvalues of the system by using the input, $u = -Kx + k_r r$.



Problem: Requires full access to the state vector, $u = -Kx$

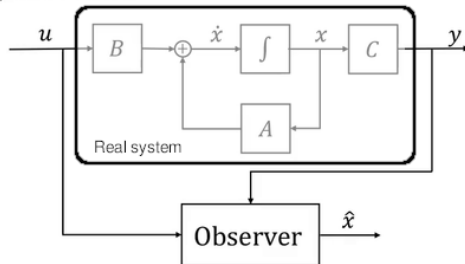
State estimation

Idea of state estimation: Develop an observer of the dynamic system that provides an estimate, $\hat{x}$, of the system's states.



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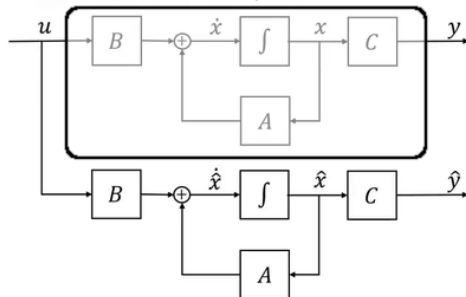
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State estimation – Open loop estimator

Goal of state estimation: $\tilde{x} = x - \hat{x} = 0$

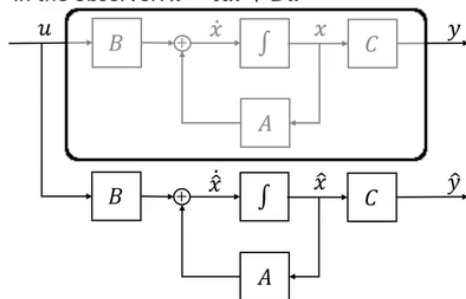
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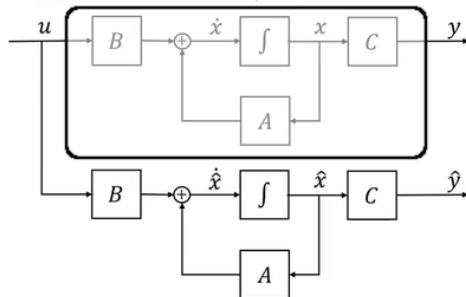
Analyze the error dynamics:

$$\begin{aligned}\dot{\tilde{x}} &= \dot{x} - \dot{\hat{x}} = Ax + Bu - A\hat{x} - Bu \\ &= A(x - \hat{x}) = A\tilde{x}\end{aligned}$$

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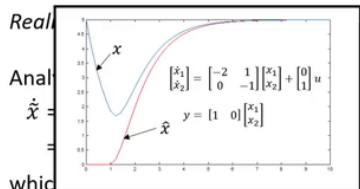
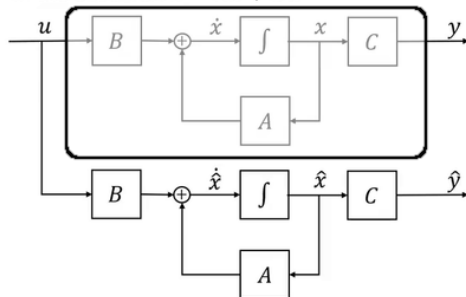
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$$\tilde{x}(t) = e^{At}\tilde{x}(0) = e^{At}(x(0) - \hat{x}(0))$$

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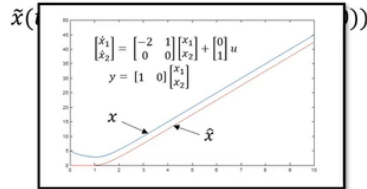
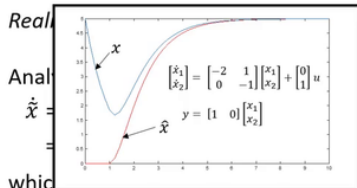
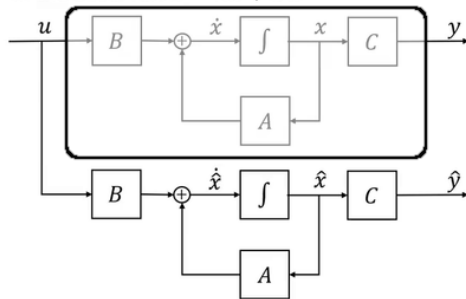


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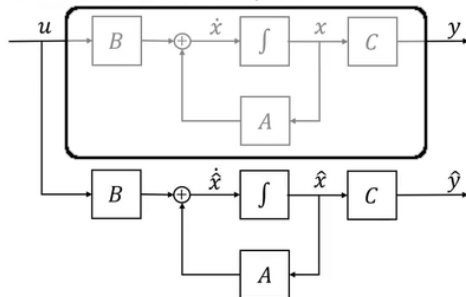
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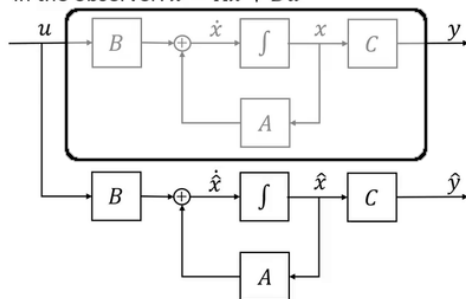
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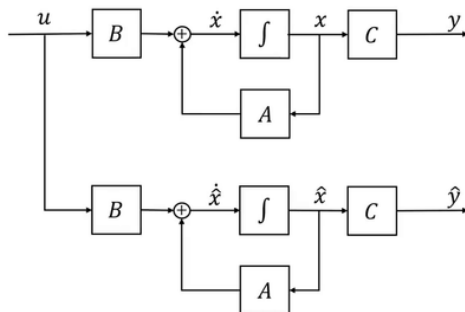
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Open loop estimation does not seem to be a good idea!

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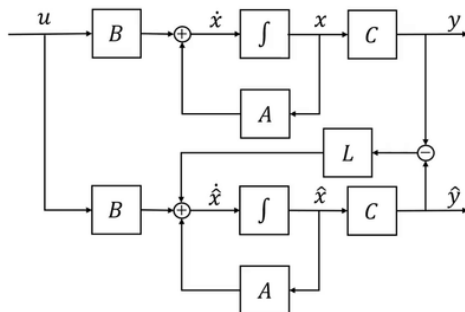
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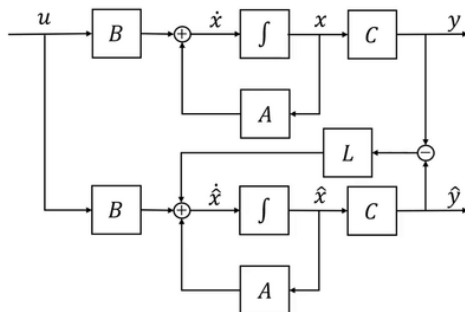
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Feed back the error to the open loop estimator via a feedback gain L :

$$\dot{\hat{x}} = A\hat{x} + Bu + L\tilde{y}$$

$$\hat{y} = C\hat{x}$$

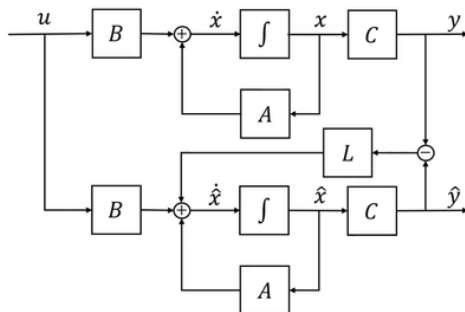
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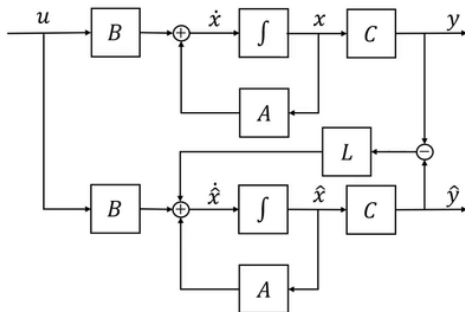
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 &= (A - LC)\tilde{x}
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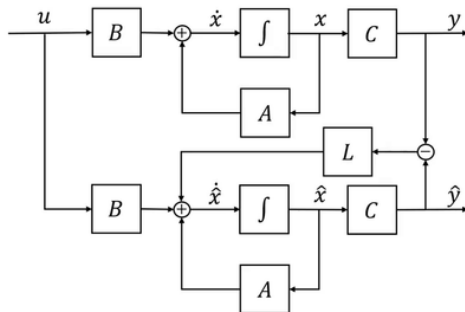
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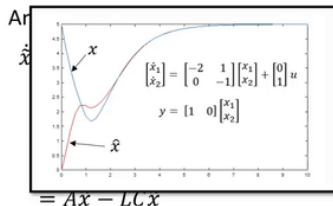
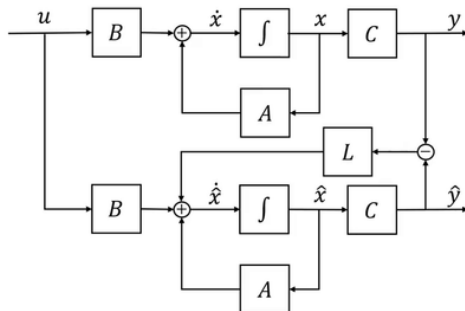
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L can be chosen such that the error dynamics converges, $\tilde{x} \rightarrow 0$ as $t \rightarrow \infty$, (if *observable*).

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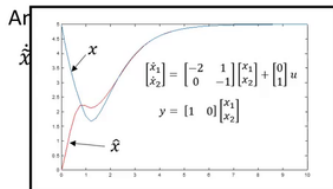
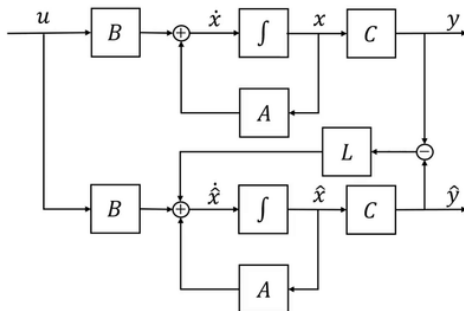
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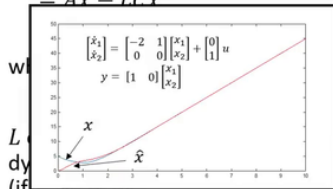
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The way to choose estimator gain is similar to that used for control design. Using the closed loop estimator, the error dynamics becomes:

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Use pole placement with a desired characteristic polynomial to choose the estimator gain, L .

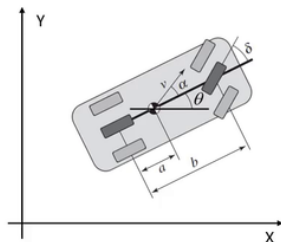
Revisit Example - Vehicle steering (Ex 7.4)

Consider the following system:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 12 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 6 \\ 3 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \end{bmatrix} u$$

where x_1 is the lateral position Y , x_2 is the heading orientation θ and u is the steering angle δ .



Vehicle data: $v_0 = 12 \text{ m/s}$

$a = 2 \text{ m}$

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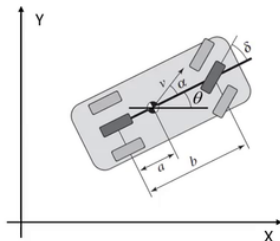
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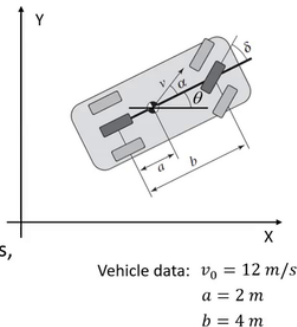
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Specification: Desired characteristic polynomial:

$$p_{des}(\lambda) = (\lambda + 6)(\lambda + 4) = \lambda^2 + 10\lambda + 24$$



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The closed loop system has the characteristic polynomial

$$\det(\lambda I - A + LC) = \begin{vmatrix} \lambda + l_1 & -12 \\ l_2 & \lambda \end{vmatrix} = \lambda^2 + l_1\lambda + 12l_2$$

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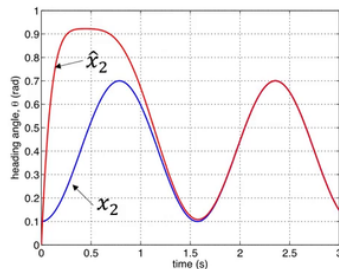
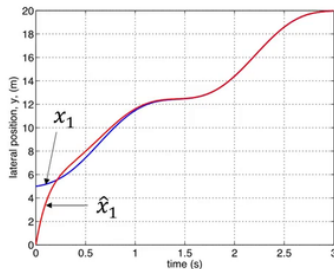
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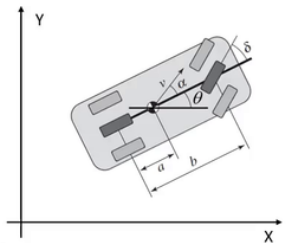
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It is not possible shape the error dynamics. We say that the system is not **observable**.

Revisit Example - Vehicle steering (Ex 7.4)

State estimator:

$$\dot{\hat{x}} = A\hat{x} + Bu + L(y - C\hat{x})$$

The closed loop estimator dynamics becomes

$$\dot{\tilde{x}} = (A - LC)\tilde{x} = \begin{bmatrix} 0 & 12 - l_1 \\ 0 & -l_2 \end{bmatrix} \begin{bmatrix} \tilde{x}_1 \\ \tilde{x}_2 \end{bmatrix}$$

The closed loop system has the characteristic polynomial unobservable

$$\det(\lambda I - A + LC) = \begin{vmatrix} \lambda & -12 + l_1 \\ 0 & \lambda + l_2 \end{vmatrix} = \lambda(\lambda + l_2)$$

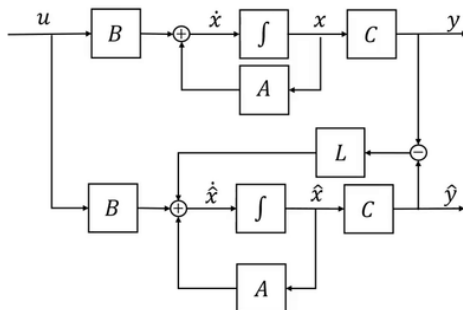
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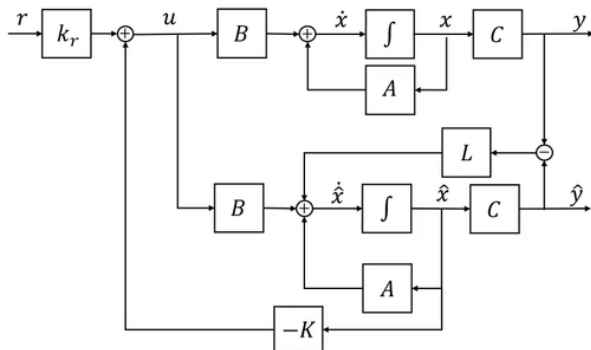
Control using estimated states

Use the estimated states for feedback, $u = -K\hat{x} + k_r r$.



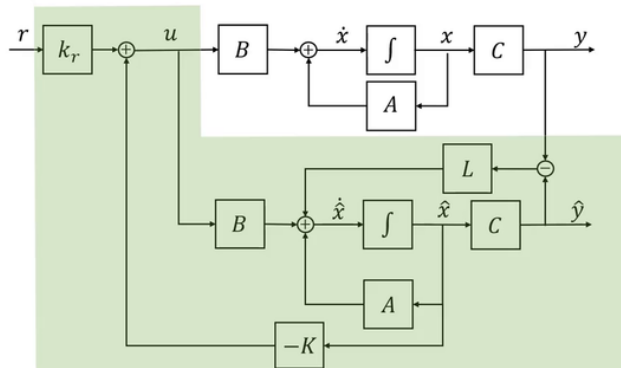
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Given the system, $\dot{x} = Ax + Bu$, $y = Cx$, the controller, $u = -K\hat{x} + k_r r$, and the state estimator, $\dot{\hat{x}} = A\hat{x} + Bu + L(y - C\hat{x})$, the closed loop system can be written as:

$$\begin{bmatrix} \dot{x} \\ \dot{\hat{x}} \end{bmatrix} = \begin{bmatrix} A - BK & BK \\ 0 & A - CL \end{bmatrix} \begin{bmatrix} x \\ \tilde{x} \end{bmatrix} + \begin{bmatrix} Bk_r \\ 0 \end{bmatrix} r$$

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Rule of thumb: Make the estimator poles 4-5 times faster than the "feedback" poles.

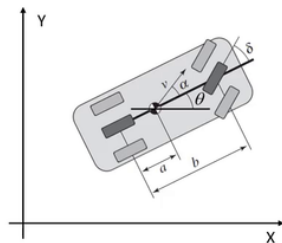
Revisit Example - Vehicle steering (Ex 7.4)

Consider the following system:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 12 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 6 \\ 3 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \end{bmatrix} u$$

where x_1 is the lateral position Y , x_2 is the heading orientation θ and u is the steering angle δ .



Vehicle data: $v_0 = 12 \text{ m/s}$

$a = 2 \text{ m}$

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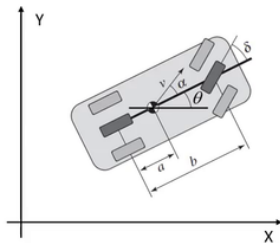
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State feedback control (poles in -1 (double pole)):

$$u = -0.0278x_1 - 0.6111x_2 + 0.0278r$$



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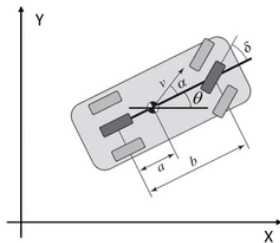
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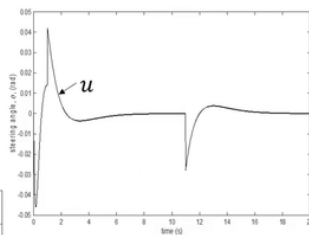
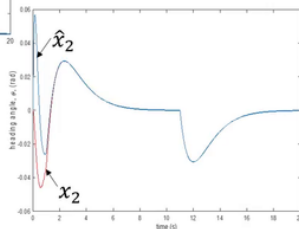
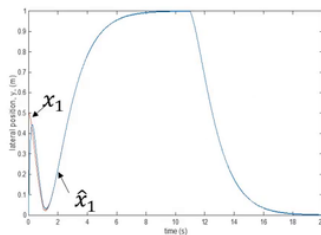
State estimator (poles in -4 and -6):

$$\begin{bmatrix} \dot{\hat{x}}_1 \\ \dot{\hat{x}}_2 \end{bmatrix} = \begin{bmatrix} 0 & 12 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \hat{x}_1 \\ \hat{x}_2 \end{bmatrix} + \begin{bmatrix} 6 \\ 3 \end{bmatrix} u + \begin{bmatrix} 10 \\ 2 \end{bmatrix} (y - \hat{x}_1)$$

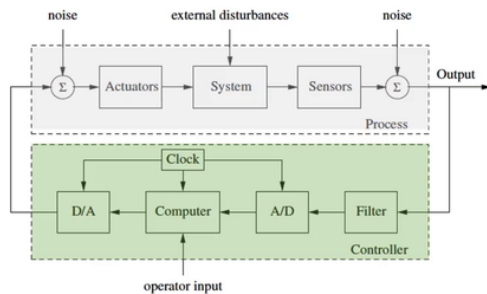


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Revisit Example - Vehicle steering (Ex 7.4)



Implementation



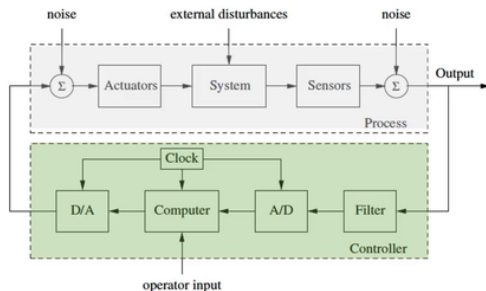
Implementation

Our controller consists of the state feedback controller,

$$u = -K\hat{x} + k_r r,$$

and the state estimator,

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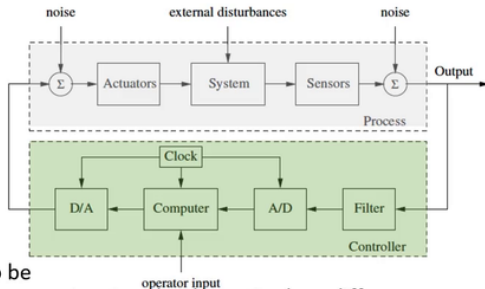
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We need to discretize the controller to be

able to implement it in a computer, by approximating the derivative by a difference:

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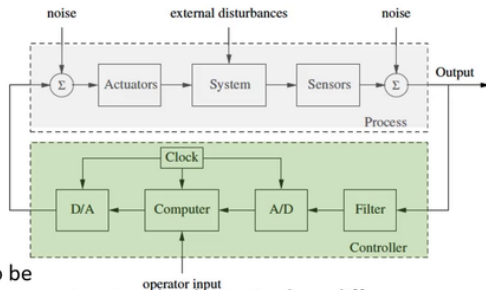
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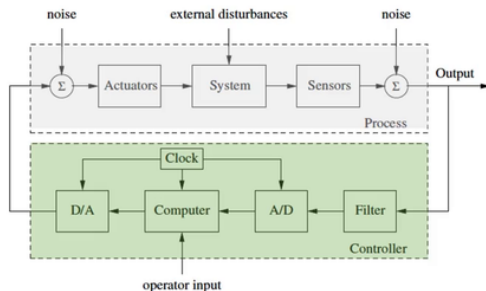
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Rewriting it as a difference equation:

$$\hat{x}(t_{k+1}) = \hat{x}(t_k) + \underbrace{(t_{k+1} - t_k)}_{h - \text{sampling time}} (A\hat{x}(t_k) + Bu(t_k) + L(y(t_k) - C\hat{x}(t_k))),$$



Implementation



In pseudocode:

```
% Control algorithm - main loop
r = adin(ch1)           % read reference
y = adin(ch2)           % get process output
u = -K*xhat + kr*r      % compute control variable
daout(ch1, u)           % set analog output
xhat = xhat + h*(A*x+B*u+L*(y-C*x)) % update state estimate
```

- Karl J. Astrom and Richard M. Murray *Feedback Systems*. Version v3.0i. Princeton University Press. September 2018. Chapter 8.